DRUGS · VERTIGO AND MÉNIÈRE'S DISEASE

Intratympanic gentamicin

Intratympanic gentamicin uses the vestibulotoxicity of the aminoglycoside on purpose. Injected through the drum, it diffuses into the inner ear and damages the vestibular hair cells (especially type I) more than the cochlea, achieving a chemical labyrinthectomy that controls vertigo in about 90% of patients with Ménière's disease. The price is a risk to hearing, which low-dose, titrated regimens have reduced.

1 CHAPTER ONE At a glance

The treatment on one page.

1.1 Summary

INTRATYMPANIC GENTAMICIN

	SUMMARY
Indication	Unilateral Ménière's disease with disabling vertigo despite medical treatment (and usually after intratympanic steroids), in a patient who accepts the risk to hearing
Preparation	Gentamicin 40 mg/ml buffered with sodium bicarbonate to about 26 to 30 mg/ml
Regimen	0.5 to 0.8 ml through the drum under topical anaesthesia, the patient lying ear-up for 30 minutes without swallowing; a single injection, repeated after 4 weeks or more only if vertigo continues (titration)
Outcomes	Vertigo control about 90%; hearing loss in roughly 10% to 30% (lower with low-dose regimens); post-treatment imbalance until compensation
Contraindications	Only hearing ear; bilateral Ménière's disease or a weak opposite labyrinth on testing; poor capacity for central compensation (elderly, cerebellar disease, visual impairment) relative

2 CHAPTER TWO Using it in ENT

Selecting and treating patients.

2.1 Before treatment

Confirm the diagnosis (definite Ménière's disease by Bárány criteria), the side, and the **function of the opposite labyrinth** (video head impulse test, calorics). Record pure-tone audiometry. Explain that the aim is to stop vertigo, not to cure the disease; that tinnitus and fullness may continue; and that the other ear may later be affected in up to a third of patients.

2.2 After treatment

Unsteadiness for days to weeks; early vestibular rehabilitation. Repeat audiometry before any further injection and 2 months after the last. Signs of effect include a reduced head impulse response and spontaneous nystagmus beating to the opposite ear.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Sensorineural hearing loss (sometimes severe), persistent imbalance and oscillopsia (particularly if the other labyrinth is weak), tinnitus, perforation and otitis media.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism: **vestibulotoxic** more than cochleotoxic; type I hair cells.
- Patient selection and the contraindications.
- Titration versus fixed-dose regimens.
- Expected outcomes for vertigo and hearing.

4.2 Common pitfalls

COMMON PITFALLS

- Treating without checking the opposite labyrinth.
- Repeating injections too soon, before the effect of the first is known.

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